Stack

(Stack, Arithmetic Expression)

Stack

- It is a linear data structure consisting of list of items.
- In stack, data elements are added or removed only at one end, called ***the top of the stack***.

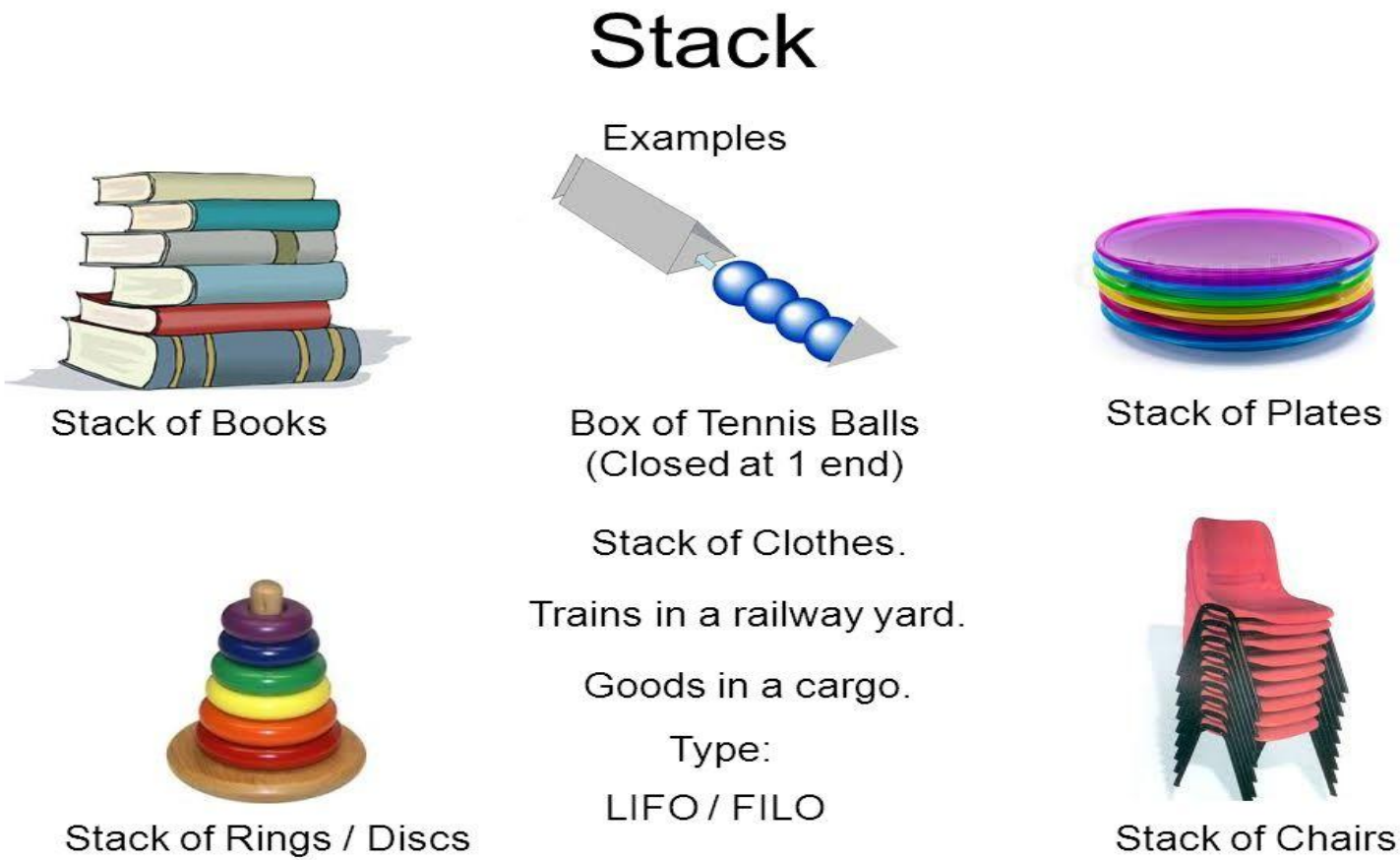


Figure: Diagrams of Stacks

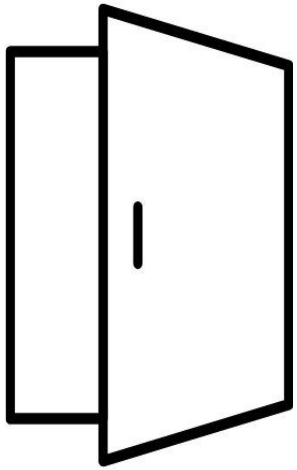
LIFO Example

LAST PERSON IN, WILL BE THE FIRST TO LEAVE

ENTRANCE
and
EXIT

last person in,
will be first out

No Back Door,
Exit from entrance



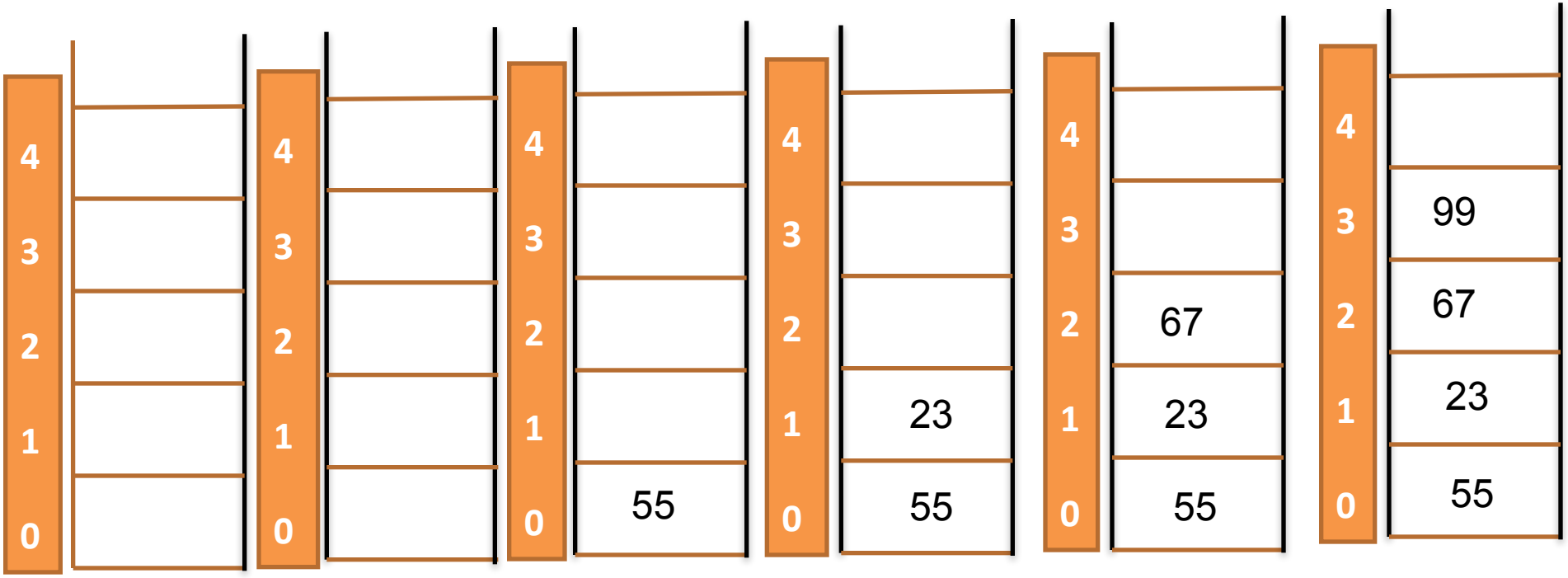
Stack Primary Operations

Two basic operations are associated with stack:

1. **“Push”** operation is used to **insert** an element into a stack.
2. **“Pop”** operation is used to **delete** an element from a stack.

Push() Example

Data to be inserted: **55, 23, 67, 99, 11**

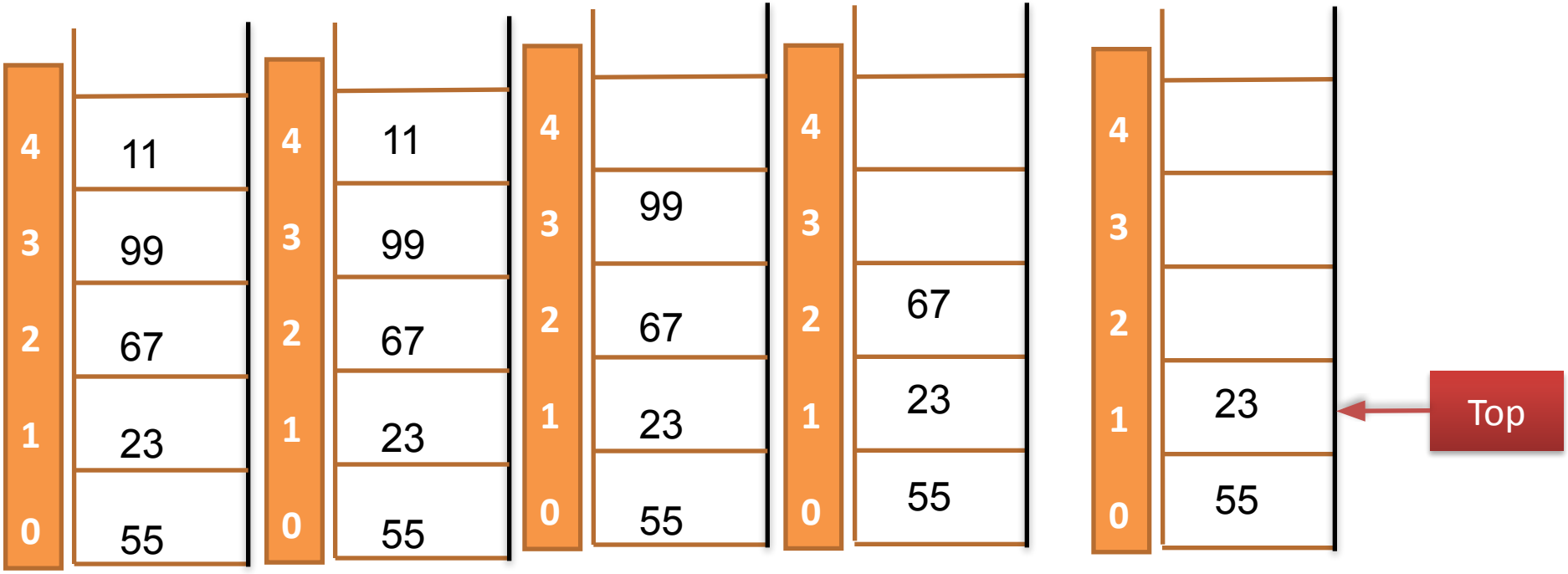


Empty stack, size()= 5	55	23	67	99	11
	push(55)	**push(23)**	**push(67)**	**push(99)**	**push(11)**
Top=-1	**Top = 0**	**Top = 1**	**Top = 2**	**Top = 3**	**Top = 4**

Next push(100) ? 100 overflow

Pop() Example

Data to be deleted: **11,99,67 and 55**



stack
size()= 5

pop(11)

Top= 4

pop(99)

Top = 3

pop(67)

Top = 2

pop(55)

Top = 1

Stack [Top] = 23 ≠ 55

Operation will not be performed

Algorithms for Push and Pop Operations

For Push Operation

Push-Stack(Stack, Top, MaxSTK, Item)

Stack Place where to store data.

Top In which location the data is to be inserted.

MaxSTK Maximum size of the stack

Item New item to be added.

1. If Top = MaxSTK then Print: Overflow and Return. /*…Stack already filled..*/
2. Set Top := Top +1
3. Set Stack[Top] := Item
4. Return.

For Pop Operation

Pop-Stack(Stack, Top, Item)

Stack Place where to store data.

Top Represents from which location the data is to be removed

Item Top element is assigned to Item.

1.If Top = -1 then Print: Underflow and Return. /*…Stack already Empty..*/

2.Set Item := Stack[Top]

3.Set Top := Top - 1

4. Return.

Stack Secondary Operations

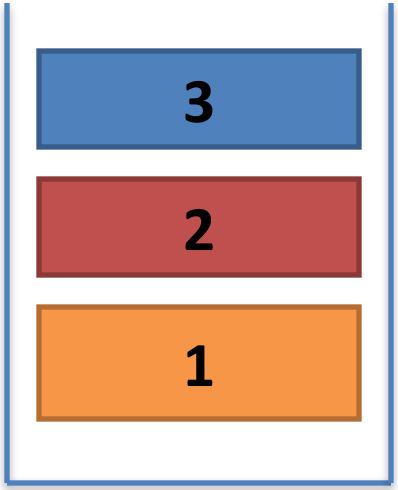
- Top():

-returns the last inserted element without removing it.

Stack Secondary Operations

- Size():

-returns the size or number of elements of a stack.

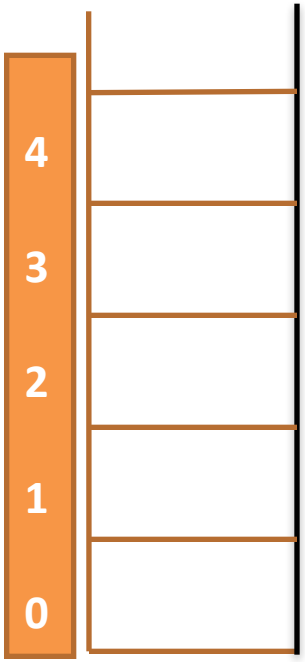


Size = 3

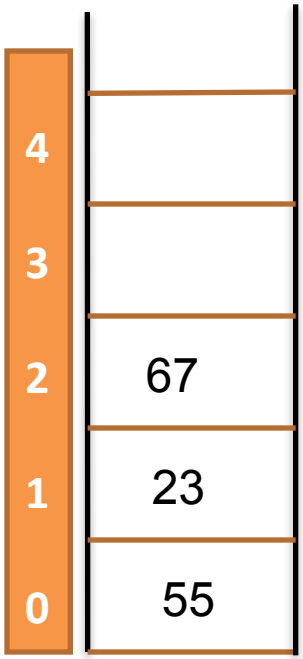
Stack Secondary Operations

- isEmpty():

- returns TRUE if the stack is Empty and FALSE otherwise.



isEmpty() ?
TRUE
if(top == -1)

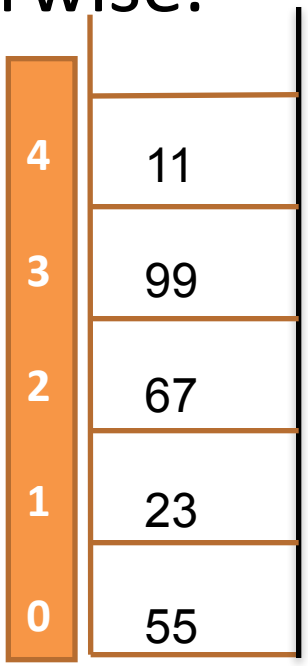


isEmpty() ?
FALSE

Stack Secondary Operations

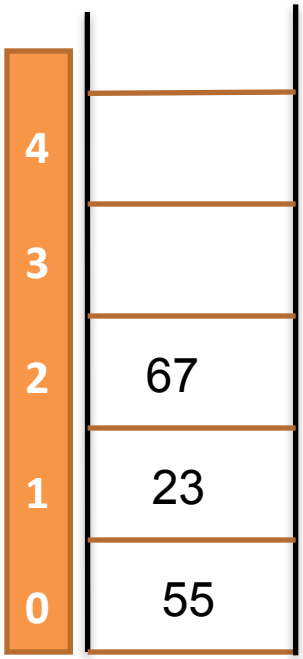
- isFull():

- returns TRUE if the stack is Full/has maximum number of elements within its size and FALSE otherwise.



Max size = 5

□isFull() ?
TRUE
if(top == maxsize-1)



□isFull() ?
FALSE

Applications of Stack

- **Arithmetic Expression Evaluation**

Calculators use a stack structure to hold values for calculation.

To calculate 7+2, you enter 7, then 2, and finally +. The calculator pushes 7 and 2 onto the stack, then pops them off when the operator + is applied, and the result 9 is pushed back onto the stack.

Stacks help in handling operator precedence and nested calculations, as they allow operations to be carried out in the correct sequence without needing parentheses.

- **Syntax Parsing**

Many compilers use a stack for parsing the syntax of expressions before translating into low level code like assembly or machine code. This is especially true for expressions written in infix notation, where the compiler needs to properly manage operator precedence and associativity to evaluate the expression correctly.

- **Solving Search Problem**

Solving a search problems, regardless of whether the approach is exhaustive or optimal, needs stack space. Example of exhaustive search methods is backtracking. Example of optimal search exploring methods is branch and bound. All of these algorithms use stacks to remember the search nodes that have been noticed but not explored yet.

- **Runtime Memory Management**

Almost all computer runtime memory environments use a special stack (the “call stack") to hold information about procedure/function calling and nesting in order to switch to the context of the called function and restore to the caller function when the calling finishes. They follow a runtime protocol between caller and callee to save arguments and return value on the stack. Stacks are an important way of supporting nested or recursive function calls.

Arithmetic Expression

- Arithmetic expression consists of operands and operators.
- Three types of arithmetic expression:

1. Infix Expression

   - Operators are placed between its two operands.

   - Example: 2 + 4, a – b / c

2. Prefix Expression(Polish Notation)

   - Operators are placed before its two operands.

   - Example: + 2 4, - a / b c

3. Postfix Expression (Reverse Polish Notation or RPN)

   - Operators are placed after its two operands.

   - Example: 2 4 +, a b c / -

Operator Precedence

- The order in which expressions are evaluated is determined by operator precedence.
- If an expression contains two or more operators with the same precedence level, the operator to the left is processed first.
- When a lower order precedence operation must be performed first, it should be surrounded by parentheses.

Operators	Precedence
( )	Highest
Unary (unary minus (-x), unary plus (+x), prefix decrement (--x), and postfix increment (x++).)	Next Highest
^	Next Highest
* , / , %	Next Highest
+ , -	Lowest

Figure: Arithmetic Operator Precedence Table

Conversion of Infix Expression into Its Equivalent Prefix Expression

1. F = A + B * C /*..Infix Expression…..*/

   = A + [* B C]

   = [+ A * B C]

   = + A * B C /*..Prefix Expression…..*/

2. F = (5 + 7) * 8 - 10 /*..Infix Expression…..*/

   = [+ 5 7] * 8 - 10

   = [* + 5 7 8] - 10

   = [- * + 5 7 8 10]

   = - * + 5 7 8 10 /*..Prefix Expression…..*/

3. F = (A + B ^ D) / (E - F)+G /*..Infix Expression…..*/

   = ?

3. F = (A + B ^ D) / (E - F) + G /*..Infix Expression…..*/

= (A + (^ B D)) / (E - F) + G

= (+ A ^ B D) / (E - F) + G

= (+ A ^ B D) / (- E F) + G

= / (+ A ^ B D) (- E F) + G

= (/ + A ^ B D - E F) + G

= + / + A ^ B D - E F G

Prefix Expression: + / + A ^ B D - E F G

Conversion of Infix Expression into Its Equivalent Postfix Expression

1. F = A + B * C /*..Infix Expression…..*/
   = A + [B C *]
   = [A B C * +]
   = A B C * + /*..Postfix Expression…..*/

2. F = (5 + 7) * 8 - 10 /*..Infix Expression…..*/
   = [5 7 +] * 8 - 10
   = [5 7 + 8 *] - 10
   = [5 7 + 8 * 10 -]
   = 5 7 + 8 * 10 - /*..Postfix Expression…..*/

3. F = (A + B ^ D) / (E - F)+G /*..Infix Expression…..*/
   = ?

Conversion of Prefix Expression into Its Equivalent Infix Expression

1. F = + A * B C /*...Prefix Expression…..*/

   = + A [B * C]

   = [A + [B * C]]

   = A + B * C /*..Infix Expression…..*/

2. F = - * + 5 7 8 10 /*..Prefix Expression…..*/

   = - * [5 + 7] 8 10

   = - [(5 + 7) * 8] 10

   = [(5 + 7) * 8] - 10]

   = (5 + 7) * 8 - 10 /*..Infix Expression…..*/

Conversion of Postfix Expression into Its Equivalent Infix Expression

1. F = A B C * + /*..Postfix Expression…..*/

   = A [B * C] +

   = [A + [B * C]]

   = A + B * C /*..Infix Expression…..*/

2. F = 5 7 + 8 * 10 - /*..Postfix Expression…..*/

   = [5 + 7] 8 * 10 -

   = [(5 + 7) * 8] 10 -

   = [[(5 + 7) * 8] - 10]

   = (5 + 7) * 8 - 10 /*..Infix Expression…..*/

Conversion of Infix Expression into Postfix Expression

Algorithm: Infix-to-Postfix (Q, P)

Here Q is an arithmetic expression in infix notation and this algorithm generates the postfix expression P using stack.

1. Scan the infix expression Q from left to right.
2. Initialize an empty stack.
3. Repeat step 4 to 5 until all characters in Q are scanned.
4. If the scanned character is an operand, add it to P.
5. If the scanned character is an operator Φ, then
   (a) If stack is empty, push Φ to the stack.
   (b) Otherwise repeatedly pop from stack and add to P each operator which has the same or higher precedence than Φ.
   (c) Push Φ to the stack.

6. If scanned character is a left parenthesis “(“, then push it to stack.

7. If scanned character is a right parenthesis “)”, then

   (a) Repeatedly pop from stack and add to P each operator until “(“ is encountered.

   (b) Remove “(“ from stack.

8. If all the characters are scanned and stack is not empty, then

   (a) Repeatedly pop the stack and add to P each operator until the stack is empty.

9. Exit.

Example: Q: 5 * (6 + 2) - 12 / 4 and P: ?

Infix Expression Q	Stack	Postfix Expression P
5		5
*	*	5
(	* (	5
6	* (	5, 6
+	* (+	5, 6
2	* (+	5, 6, 2
)	*	5, 6, 2, +
-	-	5, 6, 2, +, *
12	-	5, 6, 2, +, *, 12
/	- /	5, 6, 2, +, *, 12
4	- /	5, 6, 2, +, *, 12, 4
	-	5, 6, 2, +, *, 12, 4, /
		5, 6, 2, +, *, 12, 4, /, -

Postfix Expression P : 5, 6, 2, +, *, 12, 4, /, -

*

Example: Q: A * ((B + C) - D) / E and P: ?

Infix Expression Q	Stack	Postfix Expression P
A		A
*	*	A
(	* (	A
(	* ((	A
B	* ((	A B
+	* ((+	A B
C	* ((+	A B C
)	* (	A B C +
-	* (-	A B C +
D	* (-	A B C + D
)	*	A B C + D -
/	/	A B C + D - *
E	/	A B C + D - * E
		A B C + D - * E /

Postfix Expression P : A B C + D - * E /

*

Do on your own

- Convert the following infix expression to postfix expression:

1. 8 * (5 ^ 4 + 2) - 6 ^ 2 / (9 * 3)

2. A + (B * C - (D / E ^ F) * G) * H

Postfix Expression Evaluation

Algorithm: Postfix-Evaluation (P, Value)

Here P is an arithmetic expression in postfix notation and this algorithm finds the value of this expression using stack.

1. Scan the postfix expression P from left to right.
2. Initialize an empty stack.
3. Repeat step 4 to 5 until all characters in P are scanned.
4. If the scanned character is an operand, push it to the stack.
5. If the scanned character is an operator Φ, then
   (a) Remove two top elements of stack where A is the top element and B is the next-to-top element.
   (b) Evaluate T = B Φ A and push T to the stack.
6. Pop the stack and assign the top element of the stack to Value.
7. Exit

Example: **P : 5, 6, 2, +, *, 12, 4, /, -** and **Value: ?**

Postfix Expression P	Stack
5	5
6	5, 6
2	5, 6, 2
+	5, 8
*	40
12	40, 12
4	40, 12, 4
/	40, 3
-	37

Value: 37

END